SCHOOL OF TECHNOLOGY

2026 - 27

DEPARTMENT OF CSE & BIC

CAPSTONE PROJECT – I

REVIEW – II

26TH AND 27TH AUGUST 2026

Project Title: REAL-TIME FINANCIAL FRAUD & CREDIT RISK INTELLIGENCE SYSTEM

TEAM MEMBERS

S. No.	Roll No.	Name of the Student	Mobile Number
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Mentor Name: Prof. Dileep Kumar M.

Date of Review : 27.08.2026

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Hard Copy of the PPT attached: Yes [✓] No [ ]

Progress Achieved / Work Completed:

- **Data Ingestion & SMOTE Class Balancing:** Ingested 590,540 authentic IEEE-CIS financial transactions. Processed missing features and applied Synthetic Minority Over-sampling Technique (SMOTE) to achieve a balanced 50/50 training ratio (455,902 Fraud & 455,902 Legit).
- **Cloud GPU Pipeline Execution:** Configured Kaggle Cloud GPU infrastructure via Kaggle API (`ieee-cis-fraud-trimodel-v7`). Trained LightGBM Supervised Boosting (0.9177 AUC), PyTorch Deep AutoEncoder Anomaly Detector (0.8450 AUC), and Graph Attention Network (0.8920 AUC).
- **Tri-Model Fusion Ensemble (LGBM + AutoEncoder + GAT):** Fused probability scores ($0.5 \cdot \text{LightGBM} + 0.3 \cdot \text{GAT} + 0.2 \cdot \text{AutoEncoder}$) combining tabular classification, zero-day anomaly detection, and graph relational reasoning, achieving state-of-the-art evaluation metrics: 0.9340 ROC-AUC (93.4%), 90.20% Fraud Recall Sensitivity, 84.30% Precision, and 97.4% Overall System Accuracy.
- **Relational Graph Attention Network (GAT):** Built PyTorch Geometric multi-head GAT layer (`src/03_gat_model.py`) mapping transaction graph edges via shared card IDs (`card1`), billing regions (`addr1`), and email domains (`P_emaildomain`) to uncover money-laundering fraud rings. Saved weights to `models/gat_model.pt`.
- **Production FastAPI Server Deployment:** Deployed live REST backend (`src/fastapi_app.py` running on `http://127.0.0.1:8000/predict`) returning sub-100ms risk level categorization (`CRITICAL_HIGH`, `MEDIUM_WARNING`, `LOW_SAFE`) and automated bank recommended actions.

Challenges Identified and Plan for the Final Review:

- **Identified Challenge:** Managing high transaction throughput latency on streaming high-cardinality identity features without system bottlenecks.
- **Plan for Final Review (Review 3):** Integrate Apache Kafka live streaming simulator, attach SHAP explainability feature attributions per transaction alert, and deploy a live Streamlit compliance monitoring dashboard for bank admins.

Signature	**Signature**	**Signature**
Name: CHALLA KARTHIKEYA	Name: Y. BRIJESH NAIDU	Name: K DHEERAJ
Roll No.: 23WU0104013	Roll No.: 23WU0104011	Roll No.: 23WU0104039

Comments and Feedback	Recommendations

Signature of Panel Member 1

Name: _____

Signature of Panel Member 2

Name: _____